

CAPITAL CITY, PA, USA

WMO: 725118

Lat: 40.217N

Lon: 76.851W

Elev: 104

StdP: 100.09

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 14751

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-9.2	-20.5	0.6	-9.6	-18.2	0.8	-7.6	11.0	0.0	9.8	-0.3	3.6	290	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	33.5	23.4	32.1	22.7	30.7	22.1	24.8	30.8	24.1	29.5	23.4	28.5	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	17.9	27.4	22.4	17.3	26.8	21.9	16.8	26.2	75.7	30.7	72.6	29.8	70.0	28.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.1	-14.4	35.7	2.6	1.8	-16.3	37.1	-17.8	38.1	-19.3	39.2	-21.2	40.5		
(5)			-15.3	26.5	2.4	0.8	-17.0	27.0	-18.4	27.4	-19.8	27.9	-21.5	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-0.5	0.7	5.4	12.1	17.5	22.2	24.7	23.9	20.1	13.5	7.4	2.2		
	DBStd	9.85	5.28	4.97	5.17	4.79	4.32	3.32	2.78	2.79	3.65	4.63	4.39	4.61		
	HDD10.0	1145	327	262	161	31	1	0	0	0	0	17	102	245		
	HDD18.3	2763	583	492	401	196	71	7	0	1	23	162	328	500		
	CDD10.0	2054	2	3	19	92	233	366	456	429	302	126	24	3		
	CDD18.3	631	0	0	0	7	44	123	198	172	75	13	0	0		
	CDH23.3	5570	0	2	7	113	398	1076	1874	1445	568	86	1	0		
	CDH26.7	1922	0	0	0	29	114	357	732	497	176	16	0	0		
(14) Wind	WSAvg	2.8	3.2	3.3	3.5	3.4	2.8	2.6	2.4	2.2	2.3	2.6	2.8	3.0		
(15) Precipitation	PrecAvg	1033	72	74	83	81	111	96	92	85	89	73	88	82		
	PrecMax	1506	204	151	155	202	247	471	247	197	380	251	183	192		
	PrecMin	764	11	14	26	11	13	2	21	24	16	1	20	20		
	PrecStd	174	44	36	32	41	56	83	52	41	73	45	41	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	20.0	24.0	30.0	32.0	33.9	35.8	34.9	33.3	28.7	22.4	18.5		
		MCWB	13.3	12.7	14.4	18.2	21.9	23.6	24.3	24.2	22.6	21.3	15.8	14.9		
	2%	DB	12.8	14.3	19.9	27.0	29.7	32.1	33.8	32.7	30.9	25.9	19.4	14.0		
		MCWB	9.2	9.0	13.0	17.2	20.7	22.7	23.8	23.2	21.8	19.3	14.4	11.0		
	5%	DB	9.1	10.4	16.8	23.6	27.7	30.3	32.1	31.1	28.6	23.3	17.2	11.1		
		MCWB	6.4	5.7	11.2	15.2	19.6	21.7	22.9	22.5	20.7	18.0	12.7	8.5		
	10%	DB	7.0	7.6	13.9	21.0	25.5	28.7	30.7	29.4	26.7	21.2	14.8	8.8		
		MCWB	4.5	4.3	8.9	13.6	18.4	20.8	22.3	21.8	19.6	16.5	10.8	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.0	16.3	20.3	23.5	24.8	26.0	25.7	24.4	22.5	17.7	15.6		
		MCDB	16.7	19.1	21.0	27.1	29.5	30.9	33.2	31.5	30.8	26.3	20.1	18.0		
	2%	WB	9.4	9.6	14.4	18.3	22.0	23.9	24.9	24.8	23.2	21.0	15.6	12.1		
		MCDB	12.1	13.7	18.8	24.4	27.5	29.8	31.4	30.1	27.7	24.4	18.5	13.6		
	5%	WB	6.9	6.7	11.9	16.7	20.8	23.0	24.1	23.9	22.3	19.2	13.5	8.8		
		MCDB	8.9	9.7	15.9	22.0	26.0	28.4	29.8	28.7	26.4	22.5	16.2	10.7		
	10%	WB	4.6	4.6	9.5	14.9	19.7	22.2	23.5	23.2	21.3	17.1	11.5	6.5		
		MCDB	6.6	7.5	13.3	19.8	24.0	27.0	28.6	27.6	25.0	20.2	13.9	8.6		
(35) Mean Daily Temperature Range	MDBR		7.8	8.7	9.8	11.1	10.7	10.4	10.3	10.0	10.3	9.9	9.3	7.5		
	5% DB	MCDBR	11.4	13.4	14.3	15.9	13.4	12.1	11.7	11.5	12.6	12.9	13.0	10.6		
		MCWBR	9.0	9.3	9.1	8.2	6.2	5.0	4.4	4.4	5.3	7.0	8.6	8.8		
	5% WB	MCDBR	10.8	11.6	12.7	13.5	11.8	10.6	10.0	9.8	9.7	10.2	10.6	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.330	0.368	0.400	0.430	0.502	0.513	0.501	0.420	0.372	0.329	0.310		
	taud		2.424	2.349	2.324	2.267	2.216	2.050	2.033	2.052	2.251	2.392	2.477	2.469		
	Ebn at Noon		849	882	882	874	852	788	775	771	816	824	823	823		
	Edh at Noon		81	101	116	131	141	167	168	161	123	96	77	73		
(44) All-Sky Solar Radiation	RadAvg		1.81	2.71	3.62	4.70	5.27	5.84	5.88	5.15	4.21	2.99	2.10	1.54		
	RadStd		0.15	0.34	0.26	0.44	0.61	0.49	0.33	0.28	0.55	0.34	0.20	0.18		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	+115
Regional (64 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+65

Nomenclature: See separate page